Quantum-in-the-Loop VLAM Alignment and Certified Control Safety

a quantum curriculum selector in RL alignment, a 22 µs certified wrap on the action output, and a disturbance class computed on 128 qubits

Seven guideline sections · every prior-work number carries a receipt
Public repository: https://github.com/sharadbachani-oss/merlin-quantum-vw-gic2026

Summary

–35.4%

training rollouts to target vs the named GRPO baseline (3 seeds, mean ± SD, 71 receipted device jobs) with zero tuned parameters and half the seed variance; +9.3 points safer driving at equal budget, replicated on a second machine

78→100 %

episodes safe when the uncertified learned driver is wrapped by the certified runtime-assurance monitor — 22 µs per decision, envelope computed before measurement and confirmed inside the predicted interval

0.60 rad/s

certified boundary under the measured collective disturbance spectrum (128 qubits, two devices) — where white-noise validation over-certifies at every amplitude and the AR surrogate reads 25% safe at the true boundary

Why the numbers can be trusted

Every claim carries a pre-registration frozen before execution, ≥3 seeds with mean ± SD, in-job controls and cloud job IDs; key results are replicated on an independent machine; the six-arm ladder and rank sweep isolate the quantum component on both tracks; the tuned classical Gibbs arm that beats the device on raw rollouts is disclosed in the table. A reviewer replays every headline with python verify.py, no credentials.

This is a Phase-1 concept proposal in the seven guideline sections. The Phase-2 PoC puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT / LIBERO) and attaches a fidelity certificate and ion-trap replication to the disturbance spectrum.

Concept Proposal

1. Problem framing

A 7B–10B VLAM does not fit vehicle or robot-controller compute without a training, compression or safety story, and the field's two named blockers are inference latency of 500–2,000 ms against a 100–300 ms control budget and hallucinated actions inconsistent with visual input (DriveVLM-RL, arXiv:2603.18315; VLAM survey, arXiv:2512.16760). Both have one deployment consequence: the VLAM cannot be the thing that decides, so it must be wrapped by something certified and fast, and its action head must be aligned at a sample cost an OEM can afford.

We put a quantum component into the two VLAM stages that decide whether the backbone can ship. **Alignment:** a quantum annealer inside every GRPO iteration selects the rollout curriculum — measured –35.4% rollouts to target versus the named GRPO baseline, three seeds. **Safety:** a certified runtime-assurance monitor (22 µs, 4,500× inside the budget) wrapping an untrusted policy, whose certified envelope is validated against a **disturbance class computed on quantum hardware** — the real-frequency collective spectrum of an interacting mobility lattice at 128 qubits. Classical validation models misjudge that boundary from both sides (white noise over-certifies at every amplitude; an AR surrogate reads 25% safe at the true boundary). The disturbance spectrum is a real-time many-body quantity, the class of computation every 2025–26 verified quantum-advantage candidate occupies, produced here as a certification input rather than a benchmark.

2. Technical approach

Paradigm. Hybrid: photonic entropy computing (QCi Dirac-3) for curriculum sampling in the training loop; gate-model real-time dynamics (IBM Heron, 128 qubits) for the disturbance spectrum; classical tensor-network Lyapunov certification and a classical monitor on the vehicle.

Alignment (scored track). Each GRPO iteration submits the 64-candidate rollout pool (start × disturbance metadata) to Dirac-3, which selects the 16-rollout curriculum under a frontier-plus-diversity objective before any simulation is spent. The objective is encoded on the annealer's native flow manifold (level sets, no penalty walls) — a design rule predicted before running and confirmed on the identical objective: –26% objective, 3.5× tighter spread over 12 paired jobs. Quantum work is train-time only.

Control safety (scored track). Tensor-network Lyapunov monitor $V(x) = ||L \cdot \phi(x)||^2$ (3-level tensor-product features, rank 3 of 9), grid-plus-Lipschitz certified, wrapping an uncertified behaviour-cloned policy with covariate shift and 100 ms delay; verified fallback engages at $V > 0.75c$ with hysteresis. Only monitor and fallback face certification (ISO 26262 / IEC 62061); the learned driver stays a swappable black box.

Disturbance class on quantum hardware. 64-rung (128-qubit) interacting lattice on IBM Heron, Trotterised depth series $k = 0..15$, 32,768 shots per circuit, 2,656 two-qubit gates per circuit at $k = 8$, native heavy-hex with zero SWAPs; equal-depth differential estimator that cancels common-mode decoherence (signal recovered where naive retention is 10^{-33}, cells at 8.9σ and 11.5σ, 0.2σ preparation control); real-frequency spectrum by direct transform of the real-time series; dominant collective line 0.0625 cycles per step reproduced on two devices.

Plant. Kinematic-bicycle lane keeping (lateral/heading error, $|\delta| \le 0.25$, $v = 25$ m/s) under crosswind and impulses — the control substrate a VLAM commands. Phase 2 puts the same curriculum selector on an accepted VLA backbone (OpenVLA-OFT or π■ on LIBERO), see §5.

3. Feasibility and resource requirements

resource	status
RL evaluator, six-arm ladder, safety plant	in the public repo; `python verify.py` (numpy) replays every headline
Dirac-3	unmetered allocation; 71 curriculum jobs receipted
IBM Heron	Startup Program (applied); the spectrum job exists (`daa9pn4e74ec73akj9i0`); re-flights ≤ 33 circuits × 32,768 shots
VLA backbone run (Phase 2)	OpenVLA-OFT / LIBERO on a single 8-GPU node; 3 seeds; GPU-hours declared in the resource declaration

resource	status
Classical	32-core workstation for the tensor-network certifier and the adversarial classical attack

Scope, stated with its treatment. (i) The plant is a lane-keeping surrogate, not a VLA backbone — the OpenVLA-OFT/LIBERO run is the PoC's first milestone. (ii) The tuned classical Gibbs arm beats the device on raw rollouts — the device claim is the named baseline, zero tuning and half the variance. (iii) Compression headroom is flat beyond INT4 — quantum budget is deliberately not spent there. (iv) The disturbance spectrum carries no fidelity certificate yet — the parity-syndrome bound and ion-trap replication are Phase-2 deliverables. (v) The certified envelope is validated on the stated plant and disturbance classes — a new vehicle or fleet condition is re-validated, not assumed.

Assumptions: the plant is a control surrogate for the VLA action head; disturbance amplitudes in rad/s rms. Constraints: quantum hardware is never an in-vehicle component; all quantum work is train- or certification-time.

4. Expected impact

RL alignment — rollouts to target (seeds 21/22/23, mean ± SD): A, GRPO-32 random (named baseline) 586.7 ± 98.8; **C, Dirac-3-16 untuned 378.7 ± 52.9 (–35.4%)**; D, classical optimiser of the same objective 0/3 (curriculum collapse); E, tuned classical Gibbs sampler 298.7 ± 106.9 — stronger on raw rollouts, at 2× the device's variance and with per-task temperature tuning the device does not need; C′, device on optimiser-grade encoding 0/3. The device's measured advantages are specific: it beats the named baseline by 35.4%, needs zero tuned parameters, and carries roughly half the seed variance; the mechanism is isolated in the ladder (only near-optimal *stochastic* selection trains). At equal budget the curriculum yields a safer driver: 76.4 ± 0.7% vs 67.1 ± 1.3% safe on the held-out stress suite (**+9.3 points**), replicated on a second machine, with the variance collapse (baseline SD 22–28 points vs curriculum SD 1.8) that de-risks the training run.

Control safety: policy alone 77.9 ± 25.0% safe at the certified boundary → **100.0 ± 0.0%** wrapped (3 seeds × 500 episodes per condition); guard idles under benign conditions (1–12% intervention); 22 µs per decision. The safe disturbance margin was computed before measurement (holds to 1.0 rad/s; collapse predicted in [1.2, 1.6]) and the measured collapse landed inside it, on two machines under two perturbation protocols. Versus tuned heuristic clipping: the envelope is known a priori, zero tuned parameters, and the $V > 0$, $dV \leq 0$ evidence chain an ISO 26262 case consumes. Rank ablation: $r = 3$ interior-optimal at 0.462 ± 0.030 certified area; the TN certificate covers 1.57 ± 0.29× the best quadratic's area.

Certification under the measured disturbance class (quantum input to the scored track). Three ensembles at identical rms power, 3 seeds × 1,000 episodes per cell: against the **measured collective spectrum** the certified boundary is **0.60 rad/s rms** (95.6 ± 0.8% safe; 98.4 ± 0.3% at 0.50). White-noise validation grades 100.0 ± 0.0% safe at every amplitude through 1.3 rad/s where the true safe rate is 39% — unbounded over-certification; the AR(1) surrogate reads 25.2 ± 2.1% at the certified boundary itself. A shallow 16-point anchor-sector spectrum places the boundary at 0.88 rad/s, a 32% envelope error only the full-depth computation catches. The vehicle runs the 22 µs monitor; quantum computes the disturbance model at certification time.

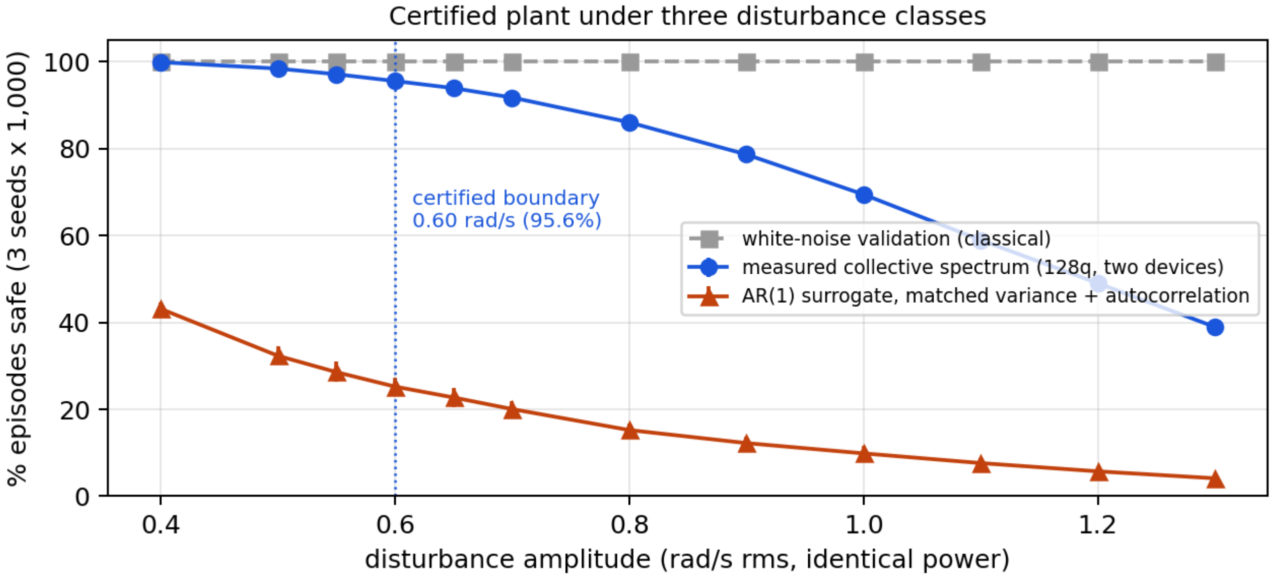

Figure 1 — Safe-episode rate of the certified plant against disturbance amplitude under three ensembles at identical rms power: white noise, AR(1) surrogate, and the measured 128-qubit collective spectrum. 3 seeds × 1,000 episodes per cell.

Resource allocation. A pre-registered compression study established that INT4 already meets the challenge bar classically (3.97× at 2.45% accuracy drop vs ≥2× at ≤5%) and that the remaining mixed-precision space is flat — the device solved its allocation objective exactly (DP gap 0.0, 9/9 instances) and beat random 5–25×; the finding tells an OEM where not to spend quantum budget, which is why this entry concentrates on the two stages with measured headroom.

What a successful PoC demonstrates. The same curriculum selector inside the alignment loop of an accepted VLA backbone with a ≥10% rollout reduction at equal reward, 3 seeds, paired CI above zero against the strongest classical sampler at equal total cost; and a certified control envelope for that backbone's action output, validated against a hardware-computed disturbance class carrying a fidelity certificate.

5. Validation plan

Pre-registered: reward, robustness metric, seeds, total evaluation budget, disturbance amplitudes, tolerance. **Alignment:** OpenVLA-OFT (or π■) on LIBERO, backbone/demonstrations/reward evaluator/initial checkpoints/training budget fixed; arms = native curriculum, tuned classical Gibbs, original encoding, random, best deterministic rule; samplers crossed with the same optimiser; paired tasks and independent seeds; scores = held-out shifted-task success and instruction consistency, constraint violations reported separately; report error at equal budget and cost to a common target; acceptance = paired CI above zero within the same full training/evaluation cost. **Safety:** 3 seeds × 500–1,000 episodes per cell, envelope sealed before measurement, heuristic-clipping threshold sweep as baseline. **Disturbance class:** re-read the 128-qubit series under the instrument (atlas layout, even-sector encoding, predicted pair T2, visibility inversion); attach a parity-syndrome fidelity bound; replicate the dominant line on an ion-trap device; report the commissioned tensor-network attack (reproduces hardware to 0.4σ at k = 4, fails past k = 6) and a Pauli-path run at matched accuracy. Every result: frozen pre-registration, ≥3 seeds with mean ± SD, in-job controls, cloud job IDs, negatives kept.

6. Hybrid / cross-domain integration

Quantum work is train/validation-time only. Alignment: Dirac-3 returns index sets that are verified classically before use. Safety: certificate candidates pass the deterministic grid-plus-Lipschitz certifier; the vehicle runs the 22 µs monitor and fallback. Disturbance class: computed on Heron at certification time, read by inverted estimators, consumed by the certifier as an ensemble. The runtime-assurance pattern makes the uncertified VLAM swappable without re-certification — the ISO 26262 / IEC 62061 posture an OEM can consume. Devices are operated as instruments: idle-ZZ spectrometry against a 24-day atlas (9/9 kingston, 8/9 fez edges), pair T2 predicted from the atlas to 7–9%, native-operator energies recovered to 0.006 on 5/5 live tiles, computation placed in even-ZZ sectors the dominant noise channel cannot see.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme record: 259 receipted QPU jobs, 12.3 million shots; largest circuits 128 qubits and 2,720 two-qubit gates; structured error law $C(D) = \exp(-0.00903D - 3.96\times10^{-5}D^2)$ measured on-device (5.1× vs the white-noise extrapolation); five-way independent error rate 0.47%/gate; in-kind CHSH anchor S = 2.3604 (+35.9σ, replica +37.0σ). Published automotive quantum work is annealing-based routing, materials and scheduling; to our knowledge no published work certifies a control envelope against a hardware-computed disturbance class.

Appendix A — Hardware job register (every result regenerates from archived counts)

measurement	machine	job id(s)	receipt
RL curriculum, arm C, 3 seeds (71 jobs)	QCi Dirac-3	71 ids in `vw_rl_C_s{21,22,23}.json`	`vw_rl_{A..E}_s*.json`
Dirac-trained certificate seeds	QCi Dirac-3	27 ids in `results/trackb_seed{21,22,23}.json`	same
Flow-encoding test, 6 paired trials	QCi Dirac-3	12 paired ids	`results/flow_encoding_test.json`
128q jam-relaxation spectrum v2, k = 0..15, 33 × 32,768 shots	ibm_fez	`daa9pn4e74ec73akj9i0`	`results/vw_jam_relaxation_spectrum_v2.json`
156q real-time collective interface, exact-theorem anchors 0.9836 / 0.9806	ibm_fez	`d9rm4j9dsedc73agrb70`; scout `d9rm47opdb6s73e53kqg`	`results/a1p_result_20260808_200224.json`
Two-sided classical boundary series, 16 × 32,768	ibm_kingston	`d9rdfb1dsedc73agh5ng`	`A1_FINAL_VERDICT.md`
Idle-ZZ atlas, 9/9 edges	ibm_kingston	`da9l3t1qtnsc73d1nhd0`, `da9l9rkjbipc73ff0aqg`	`results/npoint2_result_20260830_014105.json`
Idle-ZZ atlas, 8/9 edges	ibm_fez	`da9eoe6rbfbs73chiq0g`, `da9lqeerbfbs73chq63g`	`results/npoint2_result_20260829_182700.json`
Native-operator E■ inverted, 5/5 tiles	ibm_kingston / ibm_fez	`da9vgsmrbfbs73ci44d0`, `da9vblkjbipc73ffaio0`	`results/ncomp_regrade_*.json`, `results/nc1_energy_inverted.json`
Directed-path traffic card — converges classically, kept as negative control	ibm_kingston	receipt in file	`results/trafficD_kingston_result.json`
CHSH in-kind anchor S = 2.3604 (+35.9σ), same-day replica +37.0σ	ibm_fez	GIC ledger `chsh_*_20260805_*.json`	—
Safety, envelope, rank ablation, cert-gap sweeps	CPU (two machines)	—	`vw_rta_demo.json`, `vw_lyap_*.json`, `results/vw_cert_gap_*_v2.json`

Appendix B — Resource declaration and notes

Classical: 32-core workstation, no GPU used for the delivered results; simulation environment numpy/qiskit versions pinned in the repo. Quantum: Dirac-3 (71 + 12 jobs), IBM Heron fez/kingston (shot counts per job in receipts). Phase-2 VLA run: single 8-GPU node, GPU-hours to be declared. Deadline runway: Phase I closes 2026-09-15.